Huawei MG-ACC-46 Access Node Hardware/Environmental Fault Resolution SOP

Node: MG-ACC-46 | **Vendor:** Huawei | **Severity:** Minor | **Alarm Code:** ALARM_8928

Description

This Standard Operating Procedure (SOP) outlines the troubleshooting and resolution process for hardware/environmental faults occurring on the Huawei MG-ACC-46 Access Node at the MG Road Core Site, which are identified by the Alarm Code ALARM_8928. The MG-ACC-46 node operates as a high-capacity access node, handling a large number of user subscriptions and providing connectivity to a vast network of customers. It employs a Huawei-based access node technology stack, which relies on the NetEngine series family. The Access Node's critical functionality is ensured by the presence of redundant components, such as power supplies. However, due to the complex interaction between hardware and environmental factors, it is not uncommon for components to fail or malfunction, triggering an alarm and necessitating corrective action. In this SOP, we delve into the root causes, diagnostic procedures, and resolution steps for hardware/environmental faults, specifically focusing on ASIC memory parity errors, line card crashes, and power supply redundancy loss.

Pre-requisite Checks

- Physically verify and confirm the alarm state of the affected node (MG-ACC-46).

- Check the redundant link status (if applicable) to ascertain whether a switchover has been triggered.

- Verify the environmental conditions of the site, including temperature, humidity, and power supply stability.

- Ensure that the remote hands are dispatched to the MG Road Core site to initiate necessary physical interventions (if required).

Potential Root Causes

Cause	Probability
ASIC memory parity errors	Medium
Line card crashes	Low
Power supply redundancy loss	High

Diagnostic Commands & Expected Outputs

```
Command: display alarm, Expected Output: ALARM_8928: MG-ACC-46: Hardware/Environmental Fault, Description: This command is used to verify the alarm state of the node and confirm that the correct alarm code is triggered, indicating a hardware/environmental fault.

Command: display card, Expected Output: Card 1/0: line card crashed, Description: This command is used to diagnose potential line card
```

```
crashes and confirm whether the line card is indeed faulty.

Command: display power supply, Expected Output: Power supply 1:
failed, redundancy lost, Description: This command is used to diagnose
power supply redundancy loss and confirm whether the primary power
supply has failed.
```

Resolution Steps

Step 1: Step: Replace the faulty line card, Command: replace card 1/0, Description: Remove the faulty line card and replace it with a new one. Verify that the new line card is operational and the network is stable.

Step 2: Step: Replace the failed power supply, Command: replace power supply 1, Description: Remove the failed power supply and replace it with a new one. Verify that the new power supply is operational and the network is stable.

Post-Check Verification

```
Command: display alarm, Expected Output: No alarm triggered,
Description: Verify that the alarm state of the node is cleared and no
further alarms are triggered.

Command: display card, Expected Output: Card 1/0: operational,
Description: Verify that the replaced line card is operational and
functioning correctly.

Command: display power supply, Expected Output: Power supply 1:
operational, redundancy intact, Description: Verify that the replaced
power supply is operational and the redundancy is intact.
```

Roll-back Procedures

- Step: Restore the original line card, Command: remove card 1/0, Description: Remove the new line card and replace it with the original line card.

- Step: Restore the original power supply, Command: remove power supply 1, Description: Remove the new power supply and replace it with the original power supply.